Customer Shopping Behavior Analysis

Uncovering patterns in spending, demographics, and product preferences across **3,900 retail transactions**

PYTHON · SQL · POWER BI

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Project Overview

Python

Clean, format & transform raw data

SQL

Exploratory & advanced queries on customer segments

Power BI

Interactive dashboards for decision-making

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Dataset at a Glance

3,900

Total Records

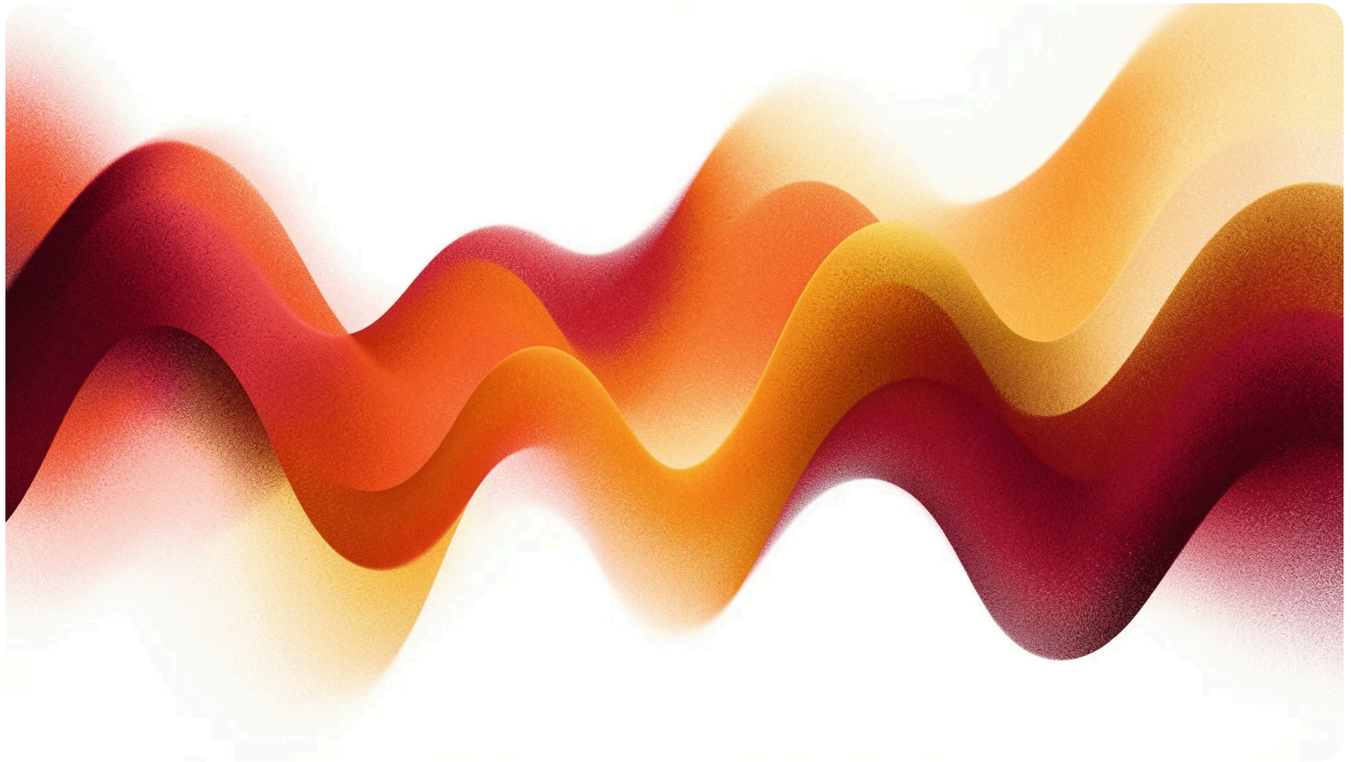
18

Columns

37

Missing Values

Review Rating field



Key Fields

- **Demographics:** Age, Gender, Location, Subscription
- **Purchases:** Item, Category, Amount, Season
- **Behavior:** Discount, Promo Code, Frequency, Rating

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EDA & Data Preparation in Python

Load & Explore

Imported via pandas; `df.info()` and `describe()` for structure

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case

Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`

Database Integration

Loaded cleaned DataFrame into MySQL for SQL analysis

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SQL Analysis: Revenue Insights

```sql
SELECT Gender,
       SUM(`Purchase Amount (USD)`) AS revenue
FROM customer
GROUP BY Gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Gender	revenue
Male	157890
Female	75191

Revenue by Gender

Compared total revenue generated by male vs. female customers.

High-Spending Discount Users

Identified customers using discounts who still spent **above average** purchase amount.

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SQL Analysis: Products & Shipping

```sql
SELECT `Item Purchased`,
       ROUND(AVG(`Review Rating`), 2) AS avg_rating
FROM customer
GROUP BY `Item Purchased`
ORDER BY avg_rating DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Item Purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

Top 5 Products by Rating

Highest average review ratings

```sql
SELECT `Shipping Type`,
       ROUND(AVG(`Purchase Amount (USD)`),2) AS avg_purchase
FROM customer
WHERE `Shipping Type` IN ('Standard','Express')
GROUP BY `Shipping Type`;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Shipping Type	avg_purchase
Express	60.48
Standard	58.46

Shipping Type Comparison

Standard vs. Express average spend

```sql
.em Purchased`,
XUND(100.0 * SUM(CASE WHEN `Discount Applied` = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS c
tomer
`Item Purchased`
discount_rate DESC
```

Filter Rows: | Export: | Wrap Cell Content:

iscount_rate
0.00
3.66
3.07
3.17
7.37

Discount-Dependent Products

Top 5 products by discount usage %

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SQL Analysis: Customers & Subscriptions

```sql
SELECT `Subscription Status`,
       COUNT(`Customer ID`) AS total_customers,
       ROUND(AVG(`Purchase Amount (USD)`),2) AS avg_spend,
       ROUND(SUM(`Purchase Amount (USD)`),2) AS total_revenue
FROM customer
GROUP BY `Subscription Status`
ORDER BY total_revenue DESC, avg_spend DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Subscription Status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status

Customer Segmentation

Classified as **New**, **Returning**, or **Loyal** based on purchase history

Repeat Buyers & Subscriptions

Customers with >5 purchases more likely to subscribe

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SQL Analysis: Categories & Age Groups

Top 3 Products per Category

Most purchased products within each category

Revenue by Age Group

Total revenue contribution per age segment

```sql
SELECT Age,
       SUM(`Purchase Amount (USD)`) AS total_revenue
FROM customer
GROUP BY Age
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Age	total_revenue
49	5552
69	5484
25	5372
41	5282
54	5282
57	5200
28	5104
19	4941
50	4930
31	4864
62	4809
32	4796

Result 20

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Power BI Dashboard

Key KPIs

Total customers, avg purchase, avg rating

Subscription Donut

High non-subscriber share — marketing opportunity

Slicers

Filter by subscription, gender, category, shipping

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Optimize Inventory

Stock Clothing & Accessories; reduce Footwear & Outerwear

Review Discount Policy

Balance sales boosts with margin control

Improve Reviews

Encourage submissions to boost credibility & organic sales

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